

Windows Server 2022 Active Directory Home Lab

Created by Juan Franquiz

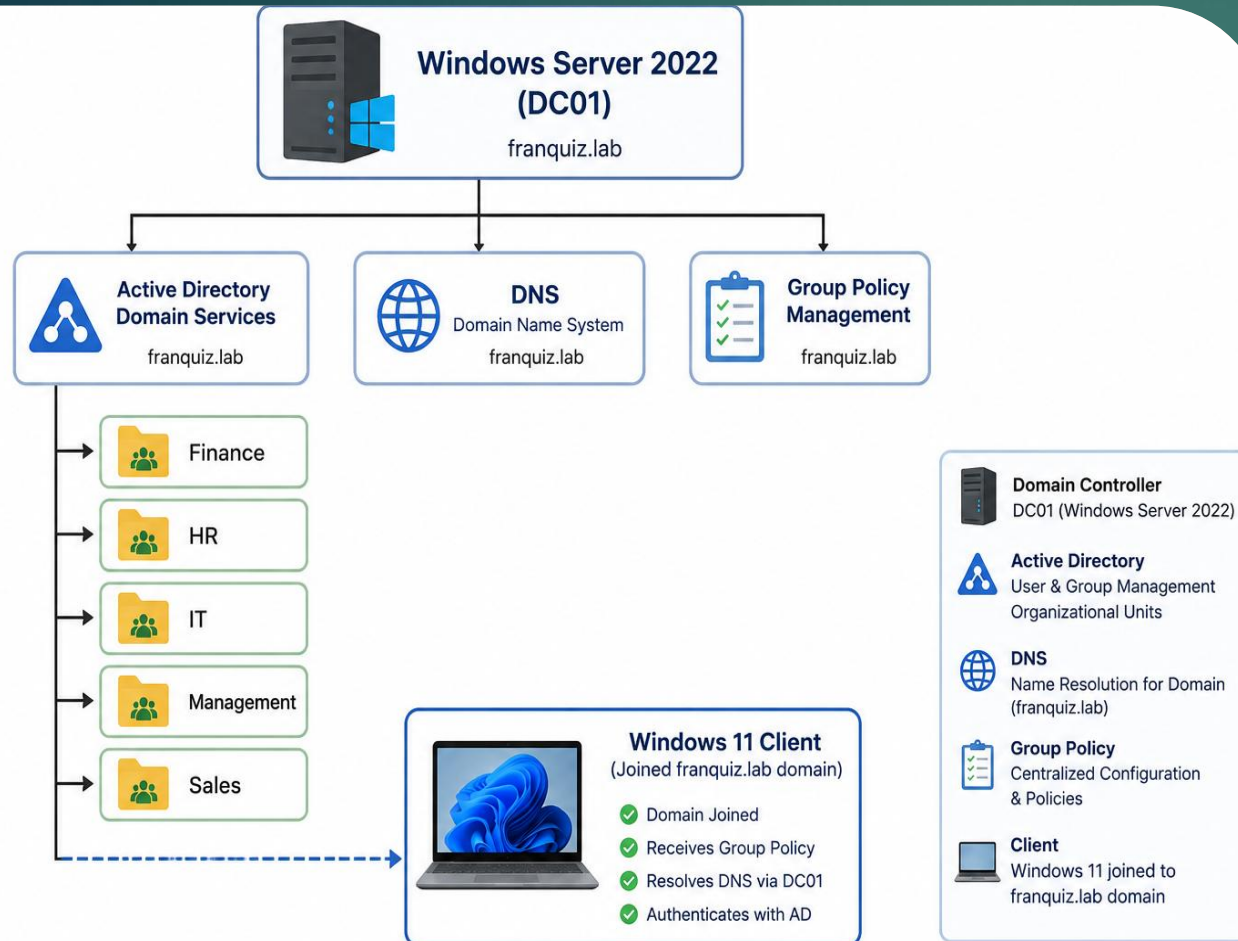
- ▶ Windows Server 2022
- ▶ Windows 11 Pro Client
- ▶ Active Directory
- ▶ DNS
- ▶ Group Policy
- ▶ SMB File Shares
- ▶ NTFS Permissions
- ▶ VirtualBox



Project Overview

Project Goal

Build and administer a Windows Server 2022 Active Directory environment to simulate a real-world enterprise network and demonstrate core IT administration skills.



Project Objectives

- Deploy Windows Server 2022 as a Domain Controller
- Configure Active Directory Domain Services (AD DS) and DNS
- Create Organizational Units (Ous), Users, and security groups
- Configure SMB file shares and NTFS permissions
- Deploy Group Policy Objects (GPOs) for centralized management
- Join Windows 11 clients to the domain

Skills Demonstrated

- Windows Server Administration
- Active directory
- DNS
- Group Policy
- User & Group Management
- File & Folder Permissions
- Windows Networking
- Troubleshooting

Windows server 2022 Domain Controller

Purpose

Deploy a Windows Server 2022 machine as the Domain controller (DC01) to provide centralized authentication, directory services, and DNS for the **franquiz.lab** domain.

Configuration

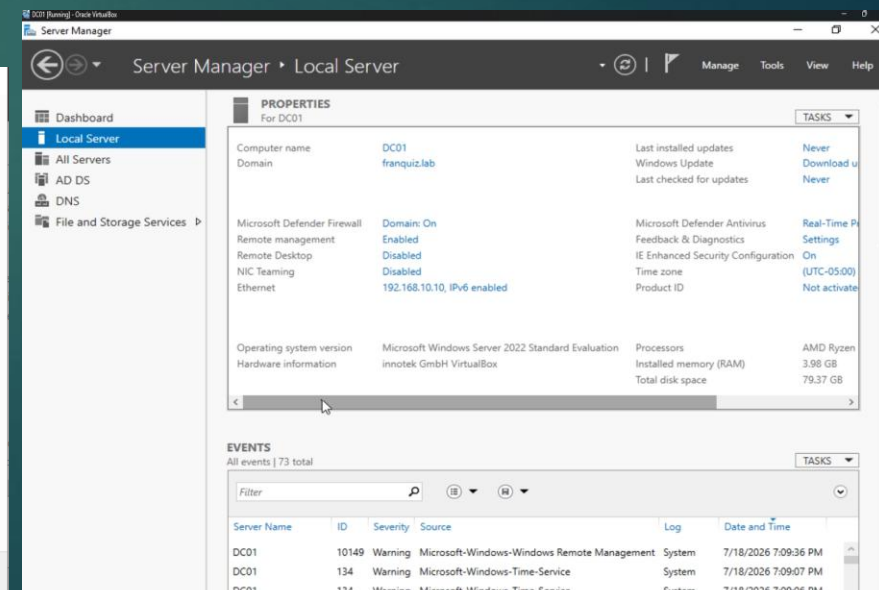
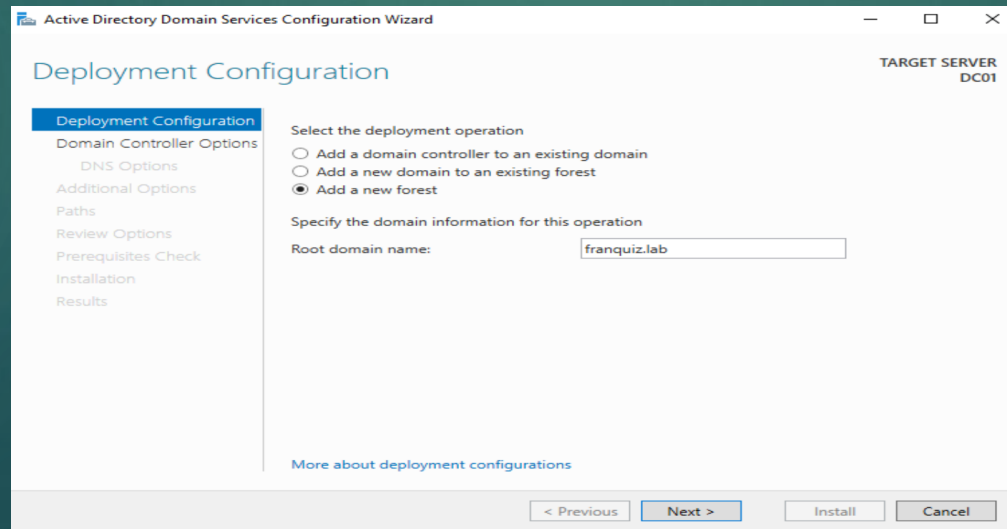
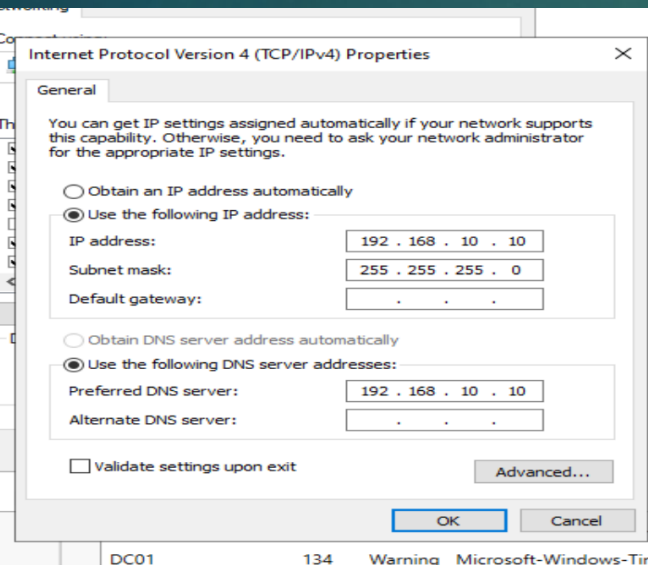
- ▶ **Server name:** DC01
- ▶ **Operating System:** Windows Server 2022 Standard
- ▶ **Domain:** franquiz.lab
- ▶ **Role:** Domain controller
- ▶ **Services Installed:** Active Directory Domain Services (AD DS), DNS

Tasks Completed

- ▶ Installed Windows Server 2022 Standard
- ▶ Renamed the server to **DC01** configured a static IPv4 address
- ▶ Installed AD DS role
- ▶ Promoted the server to a Domain controller
- ▶ Created the **franquiz.lab** forest
- ▶ Installed and configured DNS

Skills Demonstrated

- ▶ Windows Server Administration
- ▶ Active Directory Deployment
- ▶ DNS Configuration
- ▶ Server Role Installation
- ▶ Domain controller Promotion



Active Directory Unit (OU) Structure

Purpose

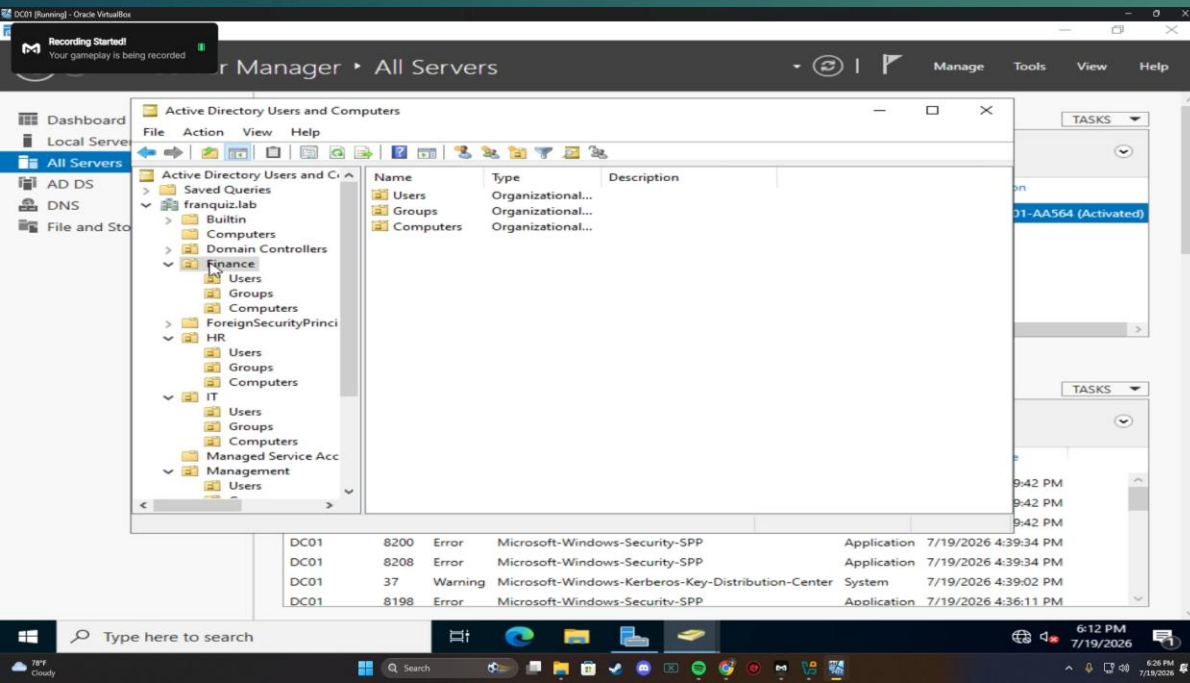
Organize users, computers, and resources into Organizational Units (OUs) to simplify administration, delegate permissions, and apply Group Policy efficiently

Skills Demonstrated

- ▶ Active Directory Administration
- ▶ Organizational Unit Design
- ▶ User & Computer Organization
- ▶ Security group Management
- ▶ Organizational Unit (OU) Management

Tasks Completed

- ▶ Created departmental Organizational Units (OUs)
- ▶ Created dedicated OUs for Users, Groups, and Computers within each department
- ▶ Organized Active directory using a scalable structure
- ▶ Prepared the environment for group Policy deployment
- ▶ Simplified user and computer management



Organizational Units Created

- ▶ Finance
- ▶ HR
- ▶ IT
- ▶ Management
- ▶ Sales

Within each department

- ▶ Users
- ▶ Groups
- ▶ Computers

User & Security Group Management

Tasks Completed

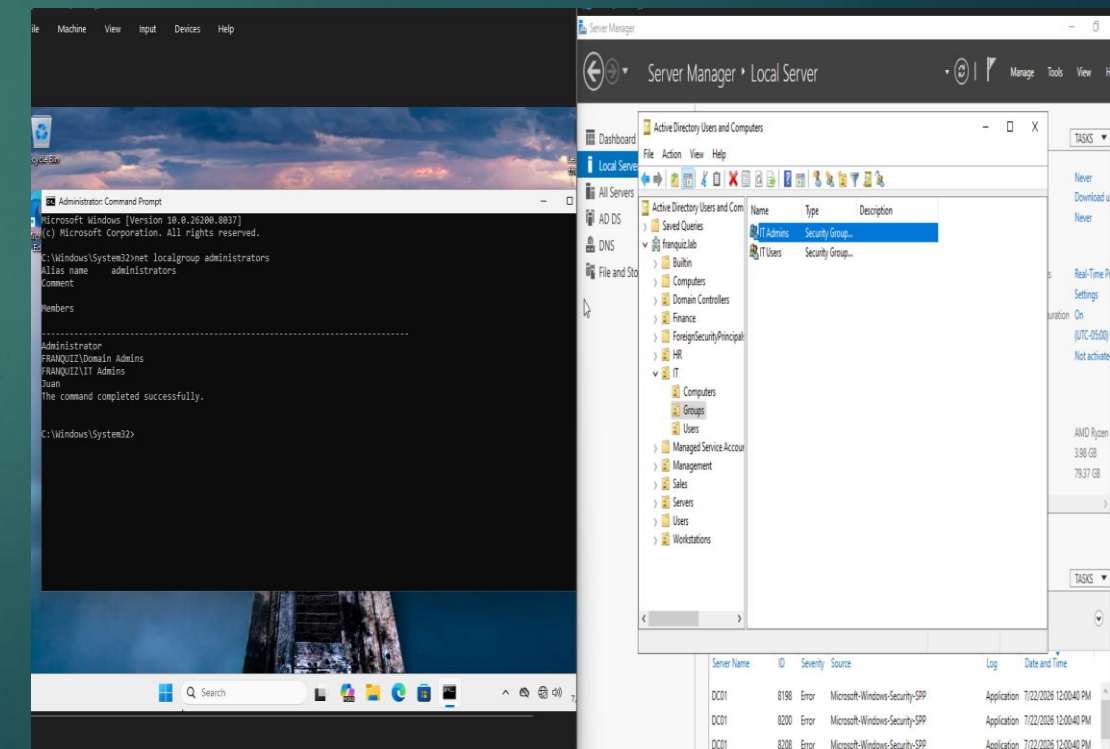
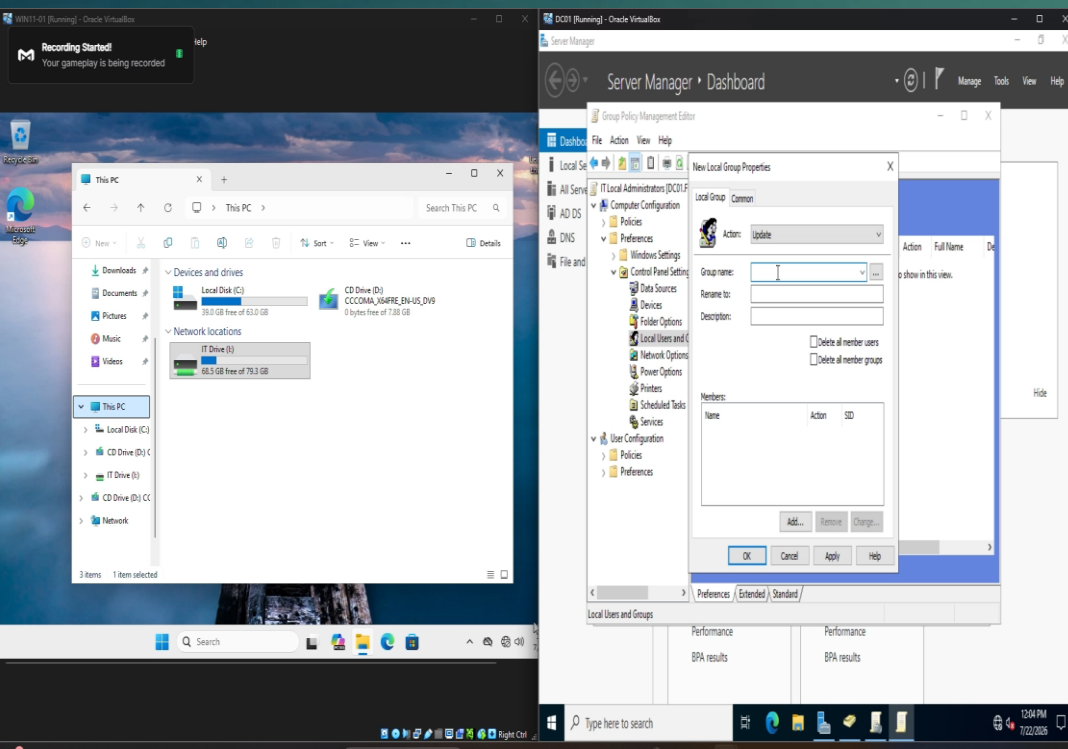
- Created domain user accounts
- Created security groups: **IT Users** and **IT Admins**
- Added Juan Franquiz to **IT Admins**
- Used RBAC so permissions are assigned to groups, not individual users
- Added the **IT Admins** domain group to the Windows 11 local Administrators group through Group Policy.

Skills Demonstrated

- Active Directory User Management
- Security Group Management
- RBAC
- Least Privilege
- Local Administrator Control
- Group Policy Administration

Verification:

The IT Admins domain group was successfully added to the Windows 11 client's local Administrators group through Group Policy.



SMB File Shares & NTFS Permissions

Purpose

Show how departmental folders were shared and secured

Tasks Completed

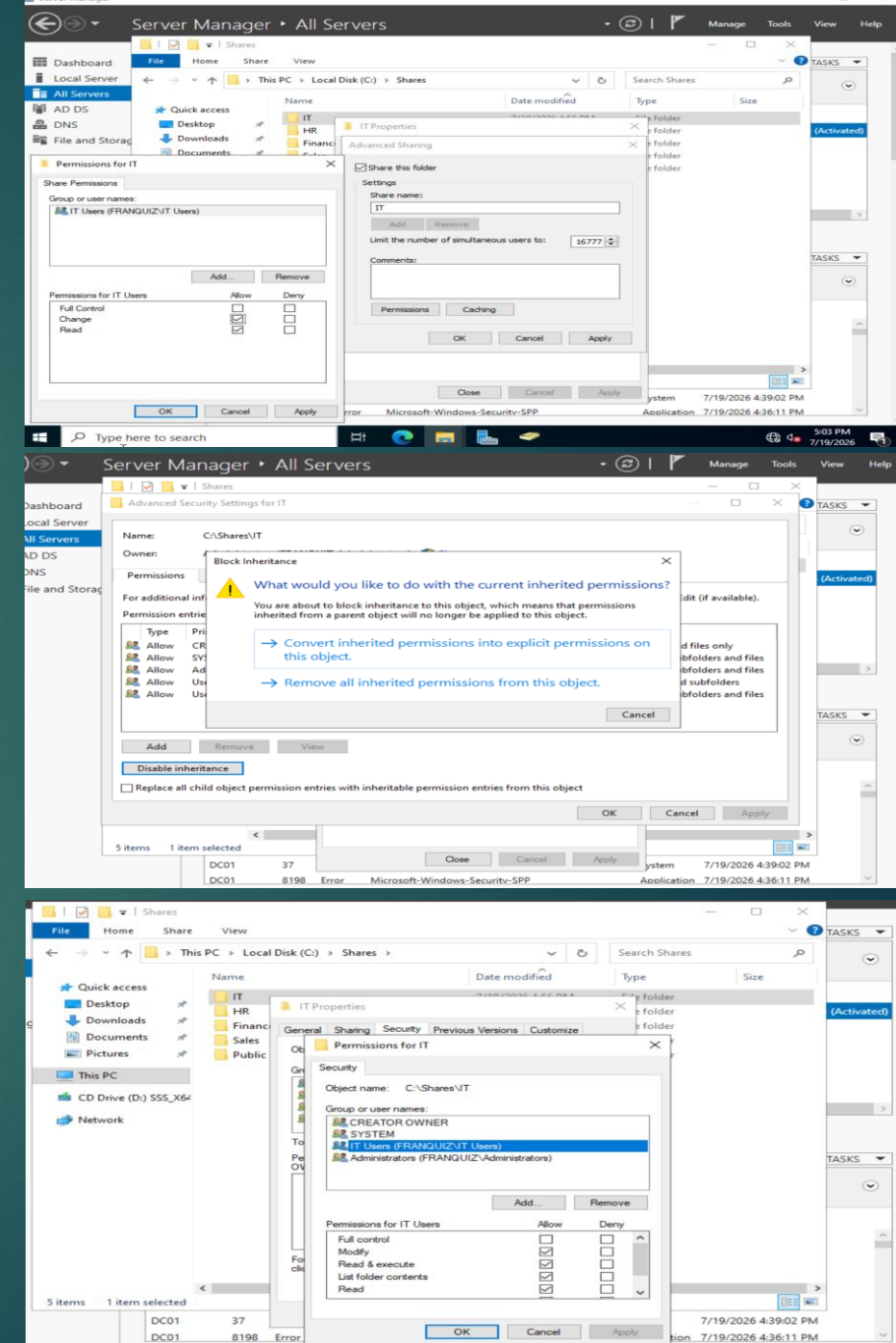
- Created **C:\Shares**
- Created department folders: IT, HR, Finance, Sales, Public
- Shared folders over the network using SMB
- Configured Share permissions
- Configured NTFS permissions
- Assigned access through security groups

Skills Demonstrated

- File Server Administration
- SMB File Sharing
- NTFS Permissions
- Share Permissions
- Least Privilege Access
- Troubleshooting Permissions

Verification:

The IT share was configured with SMB Share permissions, and NTFS permissions were assigned to the IT Users security group.



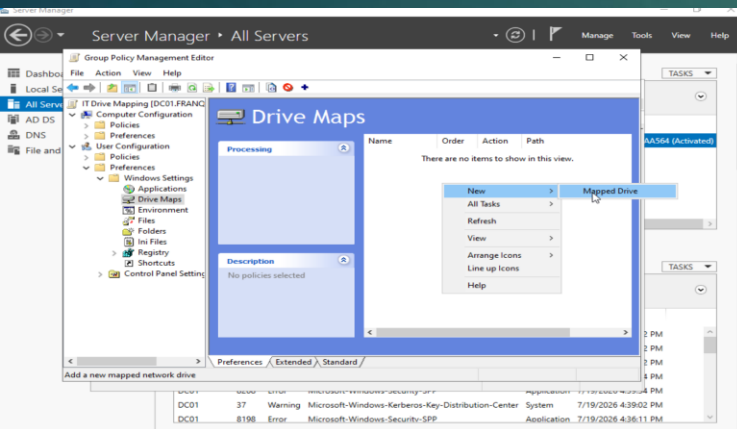
Group Policy Drive Mapping

Purpose

Show how Group Policy automatically maps the IT shared folder for users.

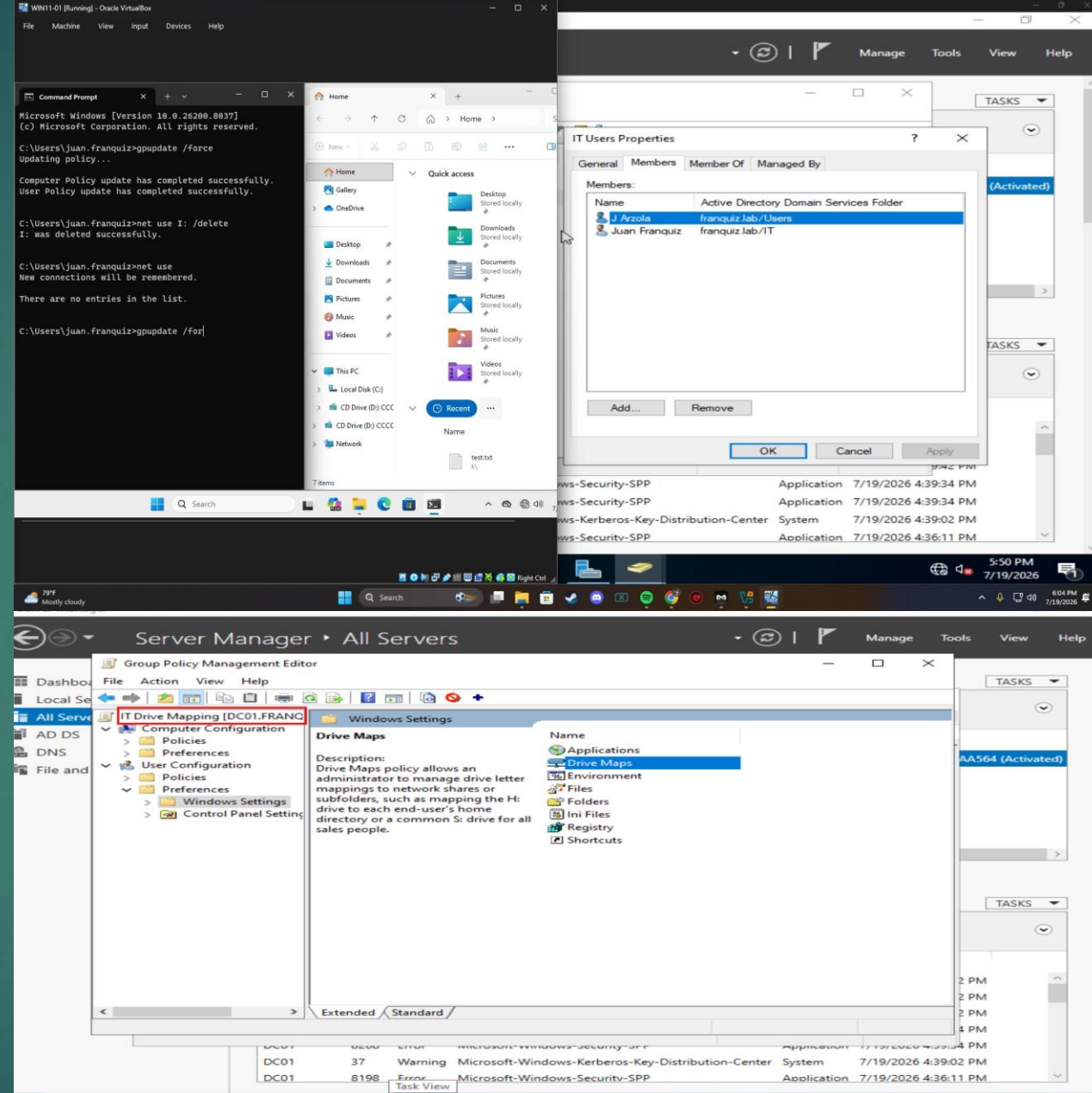
Skills Demonstrated

- Group Policy Management
- Drive Mapping
- User Configuration Policies
- SMB Share Access
- Windows Client Support
- GPO Troubleshooting



Tasks Completed

- Created an **IT Drive Mapping** GPO
- Configured Drive Maps in Group Policy Preferences
- Mapped **\\DC01\IT** to the **I:** drive
- Applied the policy into the IT OU
- Verified the drive remapped after running **gpupdate /force**



Verification:

After deleting the I: drive and refreshing Group Policy, the **\\DC01\IT** share was automatically remapped on the Windows 11 client.

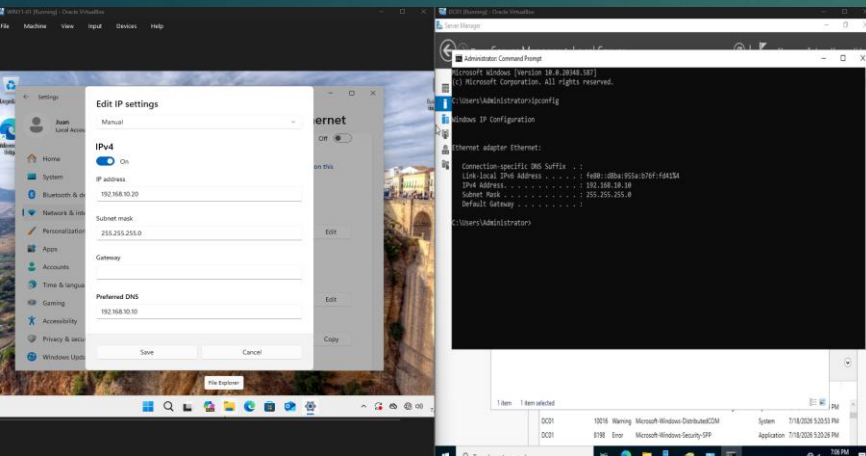
Windows 11 Domain Join & Client Testing

Purpose

Verify the Windows 11 client joined **franquiz.lab** and can communicate with DC01.

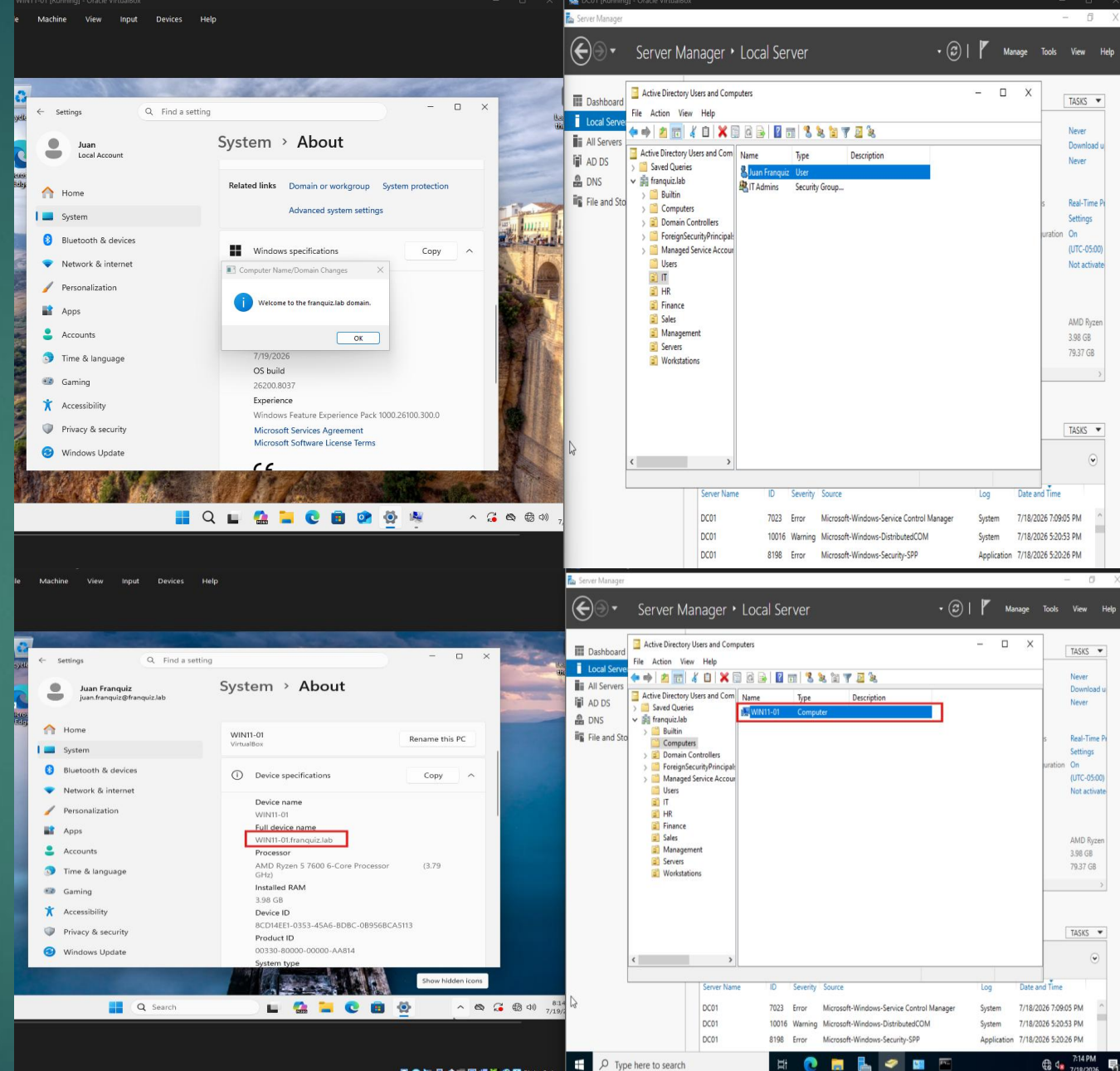
Skills Demonstrated

- Windows Client Support
- Domain Join
- DNS Configuration
- Active Directory Computer Management
- Client Troubleshooting



Tasks Completed

- Configured Windows 11 client network settings
- Set DNS to **192.168.10.10**
- Joined client to **franquiz.lab**
- Verified **WIN11-01** in Active Directory
- Confirmed domain sign-in as **juan.franquiz**



Verification:

The Windows 11 client successfully joined the **franquiz.lab** domain, appeared as **WIN11-01** in Active Directory, and used DC01 for DNS resolution.

Troubleshooting & Validation

Problem Identified

Computer Group Policy failed because the Windows 11 client clock was not synchronized with DC01.

Resolution

- Corrected the system time
- Resynchronized the client with the domain
- Ran **gpupdate /force** again

Skills Demonstrated

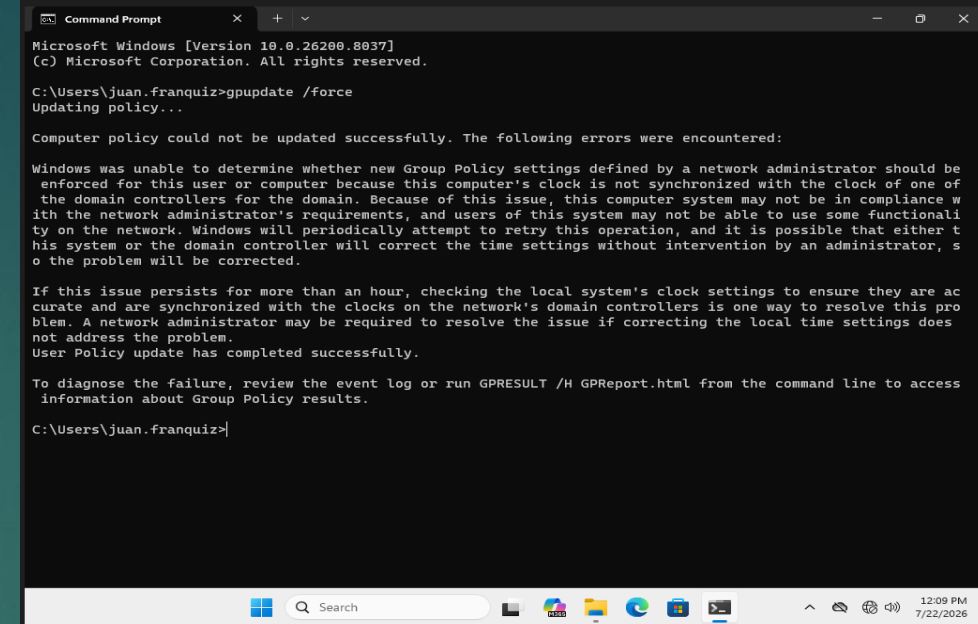
- Reading Windows error messages
- Time synchronization troubleshooting
- Group Policy troubleshooting
- Command-line validation
- Verifying applied permissions

Validation

- Computer and User Policy completed successfully
- FRANQUIZ\IT Admins appeared in the local Administrators group
- Group Policy drive mapping continued working

Verification sentence

Troubleshooting confirmed that DNS, Group Policy, domain access, and mapped drives were working correctly across the Windows Server and Windows 11 client.



```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\juan.franquiz>gpupdate /force
Updating policy...

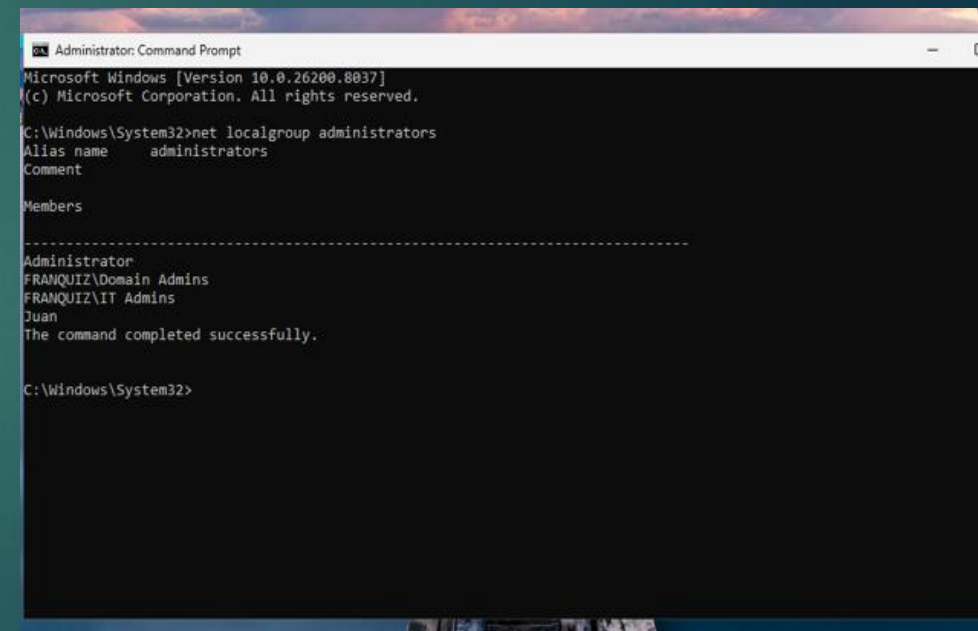
Computer policy could not be updated successfully. The following errors were encountered:

Windows was unable to determine whether new Group Policy settings defined by a network administrator should be enforced for this user or computer because this computer's clock is not synchronized with the clock of one of the domain controllers for the domain. Because of this issue, this computer system may not be in compliance with the network administrator's requirements, and users of this system may not be able to use some functionality on the network. Windows will periodically attempt to retry this operation, and it is possible that either this system or the domain controller will correct the time settings without intervention by an administrator, so the problem will be corrected.

If this issue persists for more than an hour, checking the local system's clock settings to ensure they are accurate and are synchronized with the clocks on the network's domain controllers is one way to resolve this problem. A network administrator may be required to resolve the issue if correcting the local time settings does not address the problem.
User Policy update has completed successfully.

To diagnose the failure, review the event log or run GPRESULT /H GPReport.html from the command line to access information about Group Policy results.

C:\Users\juan.franquiz>
```



```
Administrator: Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>net localgroup administrators
Alias name     administrators
Comment

Members

-----
Administrator
FRANQUIZ\Domain Admins
FRANQUIZ\IT Admins
Juan
The command completed successfully.

C:\Windows\System32>
```

Project Results & Skills Summary

Tasks Completed:

- Built Windows Server 2022 domain controller
- Created **franquiz.lab** Active Directory domain
- Organized users, groups, computers, and OUs
- Configured SMB shares and NTFS permissions
- Used Group Policy for drive mapping and local admin control
- Joined Windows 11 client to the domain
- Tested access, permissions, and troubleshooting

Skills Demonstrated:

- Active Directory Administration
- DNS Configuration
- Group Policy Management
- User & Group Management
- File Share Permissions
- Windows 11 Client Support
- Troubleshooting & Validation

Platforms

- Oracle VirtualBox
- Windows Server 2022
- Windows 11 Pro

Technologies

- Active Directory
- DNS
- SMB
- NTFS
- Group Policy
- Command-line tools

Administrative Tools

- Server Manager
- Active Directory Users and Computers
- Group Policy Management
- DNS Manager

Command-line

- gpupdate /force
- net use
- net localgroup administrators
- ipconfig
- w32tm